

Task1:

Create queries to connect to all .csv files in the folder into power query

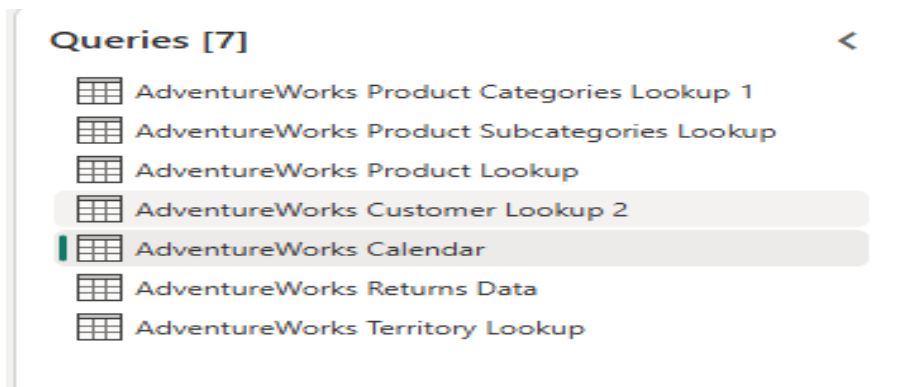
Confirm that column headers have been promoted and that all data types are correct

Add a new column to extract all characters before the dash ("" - "") in the Product SKU column, and name it “SKU Type”

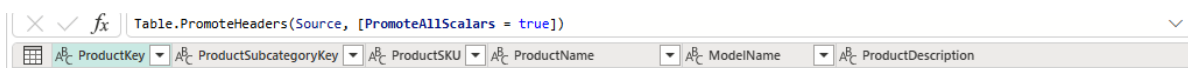
Update the SKU Type calculation above to return all characters before second dash

Replace zeros (0) in the Product Style column with “NA”

Queries to connect CSVs:



Column Header :



Data Type:



Custom Column:

fx

Table.TransformColumnTypes(Table.AddColumn("#Changed column type", "SKUType", each Text.BeforeDelimiter([ProductSKU], "-")),

	ProductColor	ProductSize	ProductStyle	ProductCost	ProductPrice	SKUType
1	Red	0	0	13.0863	34.99	HL
2	Black	0	0	12.0278	33.6442	HL
3	White	M	U	3.3963	9.5	SO
4	White	L	U	3.3963	9.5	SO
5	Blue	0	0	12.0278	33.6442	HL
6	Multi	0	U	5.7052	8.6442	CA
7	Multi	S	U	31.7244	48.0673	LJ
8	Multi	M	U	31.7244	48.0673	LJ
9	Multi	L	U	31.7244	48.0673	LJ

Update the Custom Column

fx

Table.TransformColumnTypes(Table.AddColumn("#Changed column type", "SKUType", each Text.BeforeDelimiter([ProductSKU], "-"),1)),

	ProductColor	ProductSize	ProductStyle	ProductCost	ProductPrice	SKUType
1	Red	0	0	13.0863	34.99	HL-U509
2	Black	0	0	12.0278	33.6442	HL-U509
3	White	M	U	3.3963	9.5	SO-B909
4	White	L	U	3.3963	9.5	SO-B909
5	Blue	0	0	12.0278	33.6442	HL-U509

Replace 0 with the n/a

Replace values ?

Replace one value with another in the selected columns.

Basic

Advanced

Value to find

0

Replace with

n/a

OK

Cancel

Output:

ProductColor

ProductSize

ProductStyle

Sort ascending

Sort descending

Remove empty

Text filters

Search

☒ (Select all)

☒ n/a

☒ U

☒ W

List may be incomplete.

[Load more](#)

OK

Cancel

Task-A : Implement text transformations

Duplicate the email address column and name it “Domain Name”

In the new column, remove all text/characters except for the domain name

Use transformation steps to clean up and capitalize the domain names (i.e. “Adventure Works”)

Make duplicate Column and named it

The screenshot shows a data table with columns: EmailAddress, AnnualIncome, TotalChildren, EducationLevel, Occupation, HomeOwner, and DomainName. The formula bar at the top displays: `Table.RenameColumns(#"Duplicated column", ({"EmailAddress - Copy", "DomainName"}))`. The query settings panel on the right shows the query name "AdventureWorks Custom..." and the applied steps: Source, Promoted..., Changed c..., Duplicated..., and Renamed c...

	ender	EmailAddress	AnnualIncome	TotalChildren	EducationLevel	Occupation	HomeOwner	DomainName
1		jon24@adventure-works.com	90000	2	Bachelors	Professional	Y	jon24@adventure-works.com
2		eugene10@adventure-works.com	60000	3	Bachelors	Professional	N	eugene10@adventure-works.com
3		ruben35@adventure-works.com	60000	3	Bachelors	Professional	Y	ruben35@adventure-works.com
4		christy12@adventure-works.com	70000	0	Bachelors	Professional	N	christy12@adventure-works.com
5		elizabeth5@adventure-works.co...	80000	5	Bachelors	Professional	Y	elizabeth5@adventure-works.co...
6		julio1@adventure-works.com	70000	0	Bachelors	Professional	Y	julio1@adventure-works.com
7		marco14@adventure-works.com	60000	3	Bachelors	Professional	Y	marco14@adventure-works.com
8		rob4@adventure-works.com	60000	4	Bachelors	Professional	Y	rob4@adventure-works.com
9		shannon38@adventure-works.c...	70000	0	Bachelors	Professional	N	shannon38@adventure-works.c...
10		jacquelyn20@adventure-works...	70000	0	Bachelors	Professional	N	jacquelyn20@adventure-works...

Extract the data from the duplicated column:

The screenshot shows a dialog box titled "Text between delimiters". It contains a text input field for "Start delimiter" with the value "@" and another text input field for "End delimiter" with the value "|". There are radio buttons for "Basic" (selected) and "Advanced". At the bottom, there are "OK" and "Cancel" buttons.

Text between delimiters

Enter the delimiters that mark the beginning and end of what you would like to extract.

☒ Basic ☐ Advanced

Start delimiter *

@

End delimiter *

|

OK Cancel

Replace values

Replace one value with another in the selected columns.

☒ Basic ☐ Advanced

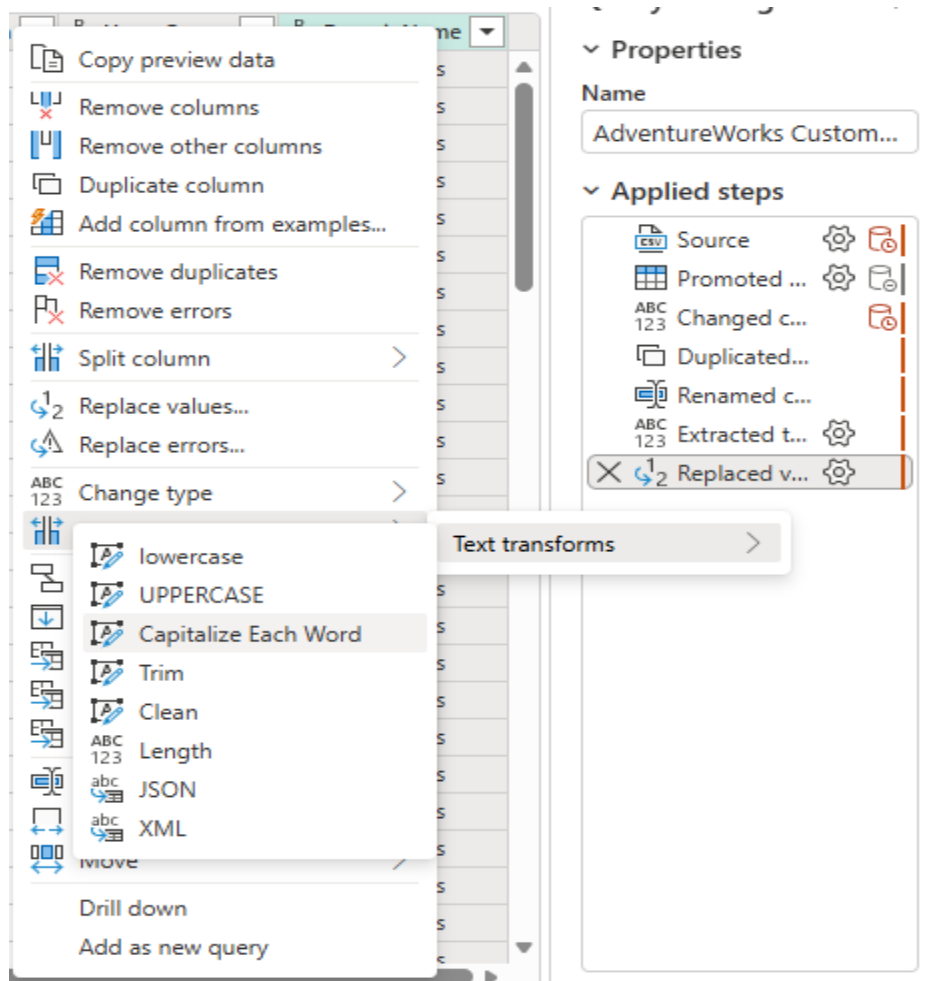
Value to find

-

Replace with

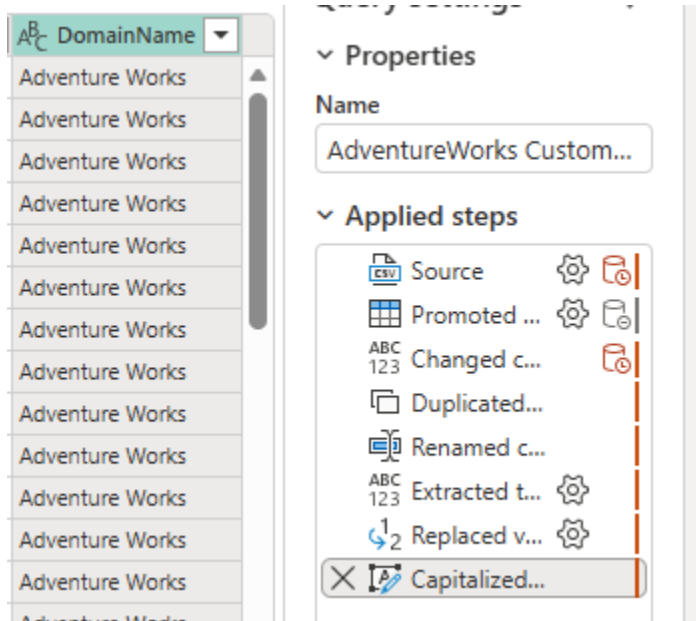
OK

Cancel



The screenshot displays a data transformation tool interface. On the left, a menu lists various operations: Copy preview data, Remove columns, Remove other columns, Duplicate column, Add column from examples..., Remove duplicates, Remove errors, Split column, Replace values..., Replace errors..., Change type, and a section for Text transforms (lowercase, UPPERCASE, Capitalize Each Word, Trim, Clean, Length, JSON, XML, and move). Below the menu are 'Drill down' and 'Add as new query' options. On the right, the 'Properties' panel shows the 'Name' as 'AdventureWorks Custom...'. The 'Applied steps' list includes: Source, Promoted..., Changed c..., Duplicated..., Renamed c..., Extracted t..., and Replaced v... (highlighted with a red bar). A 'Text transforms' panel is also visible, showing a list of text transformation options.

Output:



TASK-B: Numeric tools

What is our average product cost?

How many colors do we sell our products in?

How many distinct customers do we have?

What is the maximum annual customer income?

Return the tables to their original state

Average Cost:

✕ ✓ *f_x*

List.Average(#"Changed column type"[ProductCost])

413.6610092150164

Colours Count:

✕ ✓ *f_x*

List.NonNullCount(List.Distinct(#"Changed column type"[ProductColor]))

10

Distinct Customer:

✕ ✓ *f_x*

List.NonNullCount(List.Distinct(#"Changed column type"[CustomerKey]))

18152

Maximum Annual Income

✕

✓

fx

List.Max(#"Changed column type"[AnnualIncome])

170000

Return Original State

Queries [2]

AdventureWorks Produ...

AdventureWorks Custo...

✕

✓

fx

Table.TransformColumnTypes(#"Promoted headers", {{{"ProductKey", Int64.Type}, {"ProductSubcategoryKey", Int64.Type}, {"ProductSKU", type text}, {"ProductName", type text}, {"ModelName", type text}, {"ProductDescription", type text}, {"ProductSize", type text}, {"ProductStyle", type text}, {"ProductCost", type number}, {"ProductWeight", type number}}})

	ProductKey	ProductSubcategoryKey	ProductSKU	ProductName	ModelName	ProductDescription
1	214	31	HL-U509-R	Sport-100 Helmet, Red	Sport-100	Universal fit, well-vented, lightweight, snap-on visor.
2	215	31	HL-U509	Sport-100 Helmet, Black	Sport-100	Universal fit, well-vented, lightweight, snap-on visor.
3	218	23	SO-8909-M	Mountain Bike Socks, M	Mountain Bike Socks	Combination of natural and synthetic fibers stays dry and p
4	219	23	SO-8909-L	Mountain Bike Socks, L	Mountain Bike Socks	Combination of natural and synthetic fibers stays dry and p
5	220	31	HL-U509-B	Sport-100 Helmet, Blue	Sport-100	Universal fit, well-vented, lightweight, snap-on visor.
6	223	19	CA-1098	AWC Logo Cap	Cycling Cap	Traditional style with a flip-up brim; one-size fits all.
7	226	21	LI-0192-S	Long-Sleeve Logo Jersey, S	Long-Sleeve Logo Jersey	Unisex long-sleeve AWC logo microfiber cycling jersey
8	229	21	LI-0192-M	Long-Sleeve Logo Jersey, M	Long-Sleeve Logo Jersey	Unisex long-sleeve AWC logo microfiber cycling jersey
9	232	21	LI-0192-L	Long-Sleeve Logo Jersey, L	Long-Sleeve Logo Jersey	Unisex long-sleeve AWC logo microfiber cycling jersey
10	235	21	LI-0192-X	Long-Sleeve Logo Jersey, XL	Long-Sleeve Logo Jersey	Unisex long-sleeve AWC logo microfiber cycling jersey
11	238	14	FR-R92R-62	HL Road Frame - Red, 62	HL Road Frame	Our lightest and best quality aluminum frame made from t
12	241	14	FR-R92R-44	HL Road Frame - Red, 44	HL Road Frame	Our lightest and best quality aluminum frame made from t
13	244	14	FR-R92R-48	HL Road Frame - Red, 48	HL Road Frame	Our lightest and best quality aluminum frame made from t
14	247	14	FR-R92R-52	HL Road Frame - Red, 52	HL Road Frame	Our lightest and best quality aluminum frame made from t
15	250	14	FR-R92R-56	HL Road Frame - Red, 56	HL Road Frame	Our lightest and best quality aluminum frame made from t
16	253	14	FR-R38B-58	LL Road Frame - Black, 58	LL Road Frame	The LL Frame provides a safe comfortable ride, while offer
17	256	14	FR-R38B-60	LL Road Frame - Black, 60	LL Road Frame	The LL Frame provides a safe comfortable ride, while offer
18	259	14	FR-R38B-62	LL Road Frame - Black, 62	LL Road Frame	The LL Frame provides a safe comfortable ride, while offer
19	262	14	FR-R38R-44	LL Road Frame - Red, 44	LL Road Frame	The LL Frame provides a safe comfortable ride, while offer
20	264	14	FR-R38R-48	LL Road Frame - Red, 48	LL Road Frame	The LL Frame provides a safe comfortable ride, while offer
21	266	14	FR-R38R-52	LL Road Frame - Red, 52	LL Road Frame	The LL Frame provides a safe comfortable ride, while offer
22	268	14	FR-R38R-58	LL Road Frame - Red, 58	LL Road Frame	The LL Frame provides a safe comfortable ride, while offer

Query settings

Properties

Name

AdventureWorks Produ...

Applied steps

Source

Promoted ...

Changed c...

Snipping Tool

Queries [2]

AdventureWorks Produ...

AdventureWorks Custo...

✕

✓

fx

Table.TransformColumnTypes(#"Promoted headers", {{{"CustomerKey", type text}, {"Prefix", type text}, {"FirstName", type text}, {"LastName", type text}, {"BirthDate", type date}, {"MaritalStatus", type text}, {"Gender", type text}, {"EmailAddress", type text}, {"AnnualIncome", Int64.Type}, {"TotalChildren", Int64.Type}, {"EducationLevel", type text}, {"Occupation", type text}, {"HomeOwner", type text}}})

	CustomerKey	Prefix	FirstName	LastName	BirthDate	MaritalStatus	Gender	EmailAddress	AnnualIncome
1	11000	MR.	JON	YANG	4/8/1966	M	M	jon24@adventure-works.com	
2	11001	MR.	EUGENE	HUANG	5/14/1965	S	M	eugene10@adventure-works.com	
3	11002	MR.	RUBEN	TORRES	8/12/1965	M	M	ruben35@adventure-works.com	
4	11003	MS.	CHRISTY	ZHU	2/15/1968	S	F	christy12@adventure-works.com	
5	11004	MRS.	ELIZABETH	JOHNSON	8/8/1968	S	F	elizabeth5@adventure-works.co...	
6	11005	MR.	JULIO	RUIZ	8/5/1965	S	M	julio1@adventure-works.com	
7	11007	MR.	MARCO	MEHTA	5/9/1964	M	M	marco14@adventure-works.com	
8	11008	MRS.	ROBIN	VERHOFF	7/7/1964	S	F	rob4@adventure-works.com	
9	11009	MR.	SHANNON	CARLSON	4/1/1964	S	M	shannon38@adventure-works.c...	
10	11010	MS.	JACQUELYN	SUAREZ	2/6/1964	S	F	jacquelyn20@adventure-works...	
11	11011	MR.	CURTIS	LU	11/4/1963	M	M	curtis9@adventure-works.com	
12	11012	MRS.	LAUREN	WALKER	1/18/1968	M	F	lauren41@adventure-works.com	
13	11013	MR.	IAN	JENKINS	8/6/1968	M	M	ian47@adventure-works.com	
14	11014	MRS.	SYDNEY	BENNETT	5/9/1968	S	F	sydney23@adventure-works.com	
15	11015	MS.	CHLOE	YOUNG	2/27/1979	S	F	chloe23@adventure-works.com	
16	11016	MR.	WYATT	HILL	4/28/1979	M	M	wyatt32@adventure-works.com	
17	11017	MRS.	SHANNON	WANG	6/26/1944	S	F	shannon1@adventure-works.com	
18	11018	MR.	CLARENCE	RAI	10/9/1944	S	M	clarence32@adventure-works.c...	
19	11019	MR.	LUKE	LAL	3/7/1978	S	M	luke18@adventure-works.com	
20	11020	MR.	JORDAN	KING	9/20/1978	S	M	jordan73@adventure-works.com	
21	11021	MS.	DESTINY	WILSON	9/3/1978	S	F	destiny7@adventure-works.com	
22	11022	MR.	ETHAN	ZHANG	10/12/1978	M	M	ethan20@adventure-works.com	

Query settings

Properties

Name

AdventureWorks Custom...

Applied steps

Source

Promoted ...

Changed c...

TASK-C Date Transformations

Create a calendar table using M language

Create new columns for the below fields

Add the following columns to the calendar table:

Month Name (e.g. "January")

Month Number (e.g. "1")

Start of Year (e.g. "1/1/2020")

Year (e.g. "2020")

M-QUERY:

```
let
    StartDate = #date(2020,1,1),
    EndDate = #date(2025,12,31),

    DateList =
        List.Dates(
            StartDate,
            Duration.Days(
                EndDate - StartDate
            ) + 1,
            #duration(1,0,0,0)
        ),

    ConvertToTable =
        Table.FromList(
            DateList,
            Splitter.SplitByNothing(),
            {"Date"}
        ),

    ChangeType =
        Table.TransformColumnTypes(
            ConvertToTable,
            {"Date", type date}
        ),

    AddMonthName =
        Table.AddColumn(
```

```
        ChangeType,  
        "Month Name",  
        each Date.MonthName([Date]),  
        type text  
    ),
```

```
AddMonthNumber =  
    Table.AddColumn(  
        AddMonthName,  
        "Month Number",  
        each Date.Month([Date]),  
        Int64.Type  
    ),
```

```
AddStartOfYear =  
    Table.AddColumn(  
        AddMonthNumber,  
        "Start of Year",  
        each Date.StartOfYear([Date]),  
        type date  
    ),
```

```
AddYear =  
    Table.AddColumn(  
        AddStartOfYear,  
        "Year",  
        each Date.Year([Date]),  
        Int64.Type  
    )
```

```
in  
    AddYear
```

Output:

	 Date 	 Month Name 	¹ ₂ 3 Month Number 	 Start of Year 	¹ ₂ 3 Year 
1	1/1/2020	January	1	1/1/2020	2020
2	1/2/2020	January	1	1/1/2020	2020
3	1/3/2020	January	1	1/1/2020	2020
4	1/4/2020	January	1	1/1/2020	2020
5	1/5/2020	January	1	1/1/2020	2020
6	1/6/2020	January	1	1/1/2020	2020
7	1/7/2020	January	1	1/1/2020	2020
8	1/8/2020	January	1	1/1/2020	2020
9	1/9/2020	January	1	1/1/2020	2020
10	1/10/2020	January	1	1/1/2020	2020
11	1/11/2020	January	1	1/1/2020	2020
12	1/12/2020	January	1	1/1/2020	2020
13	1/13/2020	January	1	1/1/2020	2020
14	1/14/2020	January	1	1/1/2020	2020
15	1/15/2020	January	1	1/1/2020	2020
16	1/16/2020	January	1	1/1/2020	2020
17	1/17/2020	January	1	1/1/2020	2020
18	1/18/2020	January	1	1/1/2020	2020
19	1/19/2020	January	1	1/1/2020	2020